MIR AQSA

Srinagar, Jammu & Kashmir | 7006841224 | miraqsa186@gmail.com

PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience in machine learning, deep learning, and Python-based application development. Built predictive classification models and a CNN-based plant disease classifier, and developed a full-stack employee management system using FastAPI and Streamlit. Interested in applying AI/ML and LLM application concepts to practical products.

TECHNICAL SKILLS

Programming & APIs: Python, FastAPI, REST APIs, Streamlit, Pydantic
Machine Learning: scikit-learn, Random Forest, XGBoost, SVM, model evaluation, cross-validation
Deep Learning & Data: TensorFlow, CNNs, MobileNetV2, data preprocessing, classification, sentiment analysis
Application & Security: JWT authentication, JSON persistence, PostgreSQL (foundational), ECC, AES, ChaCha20, PQC concepts
AI/LLM Learning: Embeddings, semantic search, RAG fundamentals, LangChain concepts

EXPERIENCE & APPLIED ML WORK

Research & Development Intern – UAV Security | NIELIT • Nov 2025 – Apr 2026
- Developed a secure UAV–ground control station communication framework using ECC-based key exchange and adaptive symmetric encryption.
- Implemented dynamic switching between ChaCha20 and AES based on network load to balance performance and robustness.
- Explored post-quantum cryptography concepts for future-ready UAV security mechanisms.

Credit Score Prediction – Paisa Bazaar Simulation • Aug 2025 – Oct 2025
- Built a classification model to predict Good, Standard, and Bad credit-score categories, achieving 92% overall accuracy on simulated data.
- Compared Random Forest, XGBoost, and SVM using cross-validation; reported a 15% improvement in classification precision over baseline methods.

Sentiment Analysis – Trader Behavior & Market Analysis • Jun 2025 – Jul 2025
- Analyzed Bitcoin market sentiment alongside historical trading data, identifying a reported 35% correlation between positive sentiment shifts and trade profitability.
- Used the Fear & Greed Index and historical data to model trade direction with 78% accuracy and examine position-size changes during sentiment extremes.

PROJECTS

Talent Management System (TMS) | Python, FastAPI, Streamlit, Pydantic, JWT, JSON
- Built a full-stack employee management application with registration, login, employee details, attendance, break tracking, and leave applications.
- Designed a layered backend with routes, services, and repository modules, using JSON files for employee, attendance, and leave persistence.
- Implemented JWT-based authenticated sessions and an optional workplace-location verification step for attendance check-in.

Apple Leaf Disease Classification | TensorFlow, CNN, MobileNetV2
- Developed a deep-learning image classifier using MobileNetV2, reporting 92% accuracy for apple leaf disease classification.
- Experimented with GoogleNet and custom CNN architectures and performed image preprocessing, training, and evaluation.

EDUCATION

B.Tech, Computer Science & Engineering | University of Kashmir | 2021 – 2024 | CGPA: 8.2/10
Diploma, Computer Science & Engineering | Government Polytechnic College | 2018 – 2020